

BEVFormer: A Unified BEV Framework for Multi-Camera 3D Recognition

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Sensor Configuration: A Roadmap

Tendency:

- More sensors
- Multi-view cameras + LiDAR
- Different sensors complement each other

C+L

16



8

1

2014

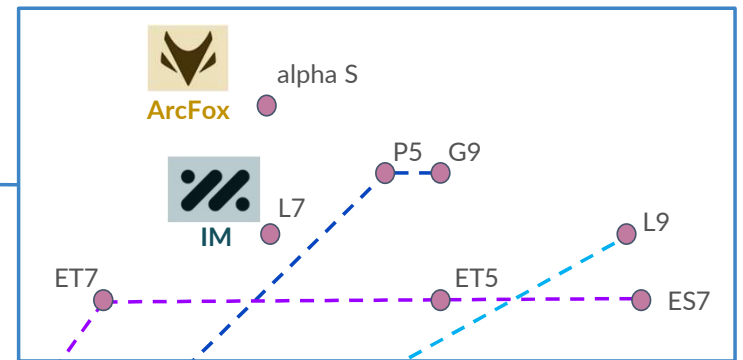
2016

2018

2020

2022

2



Tesla

HW(Hardware)1.0

HW2.0

HW2.5

HW3.0

No
RaDAR



NIO

ES8



XPeng

G3



LiAuto

ES6

EC6

One



ArcFox

alpha S



IM

L7

P5

G9

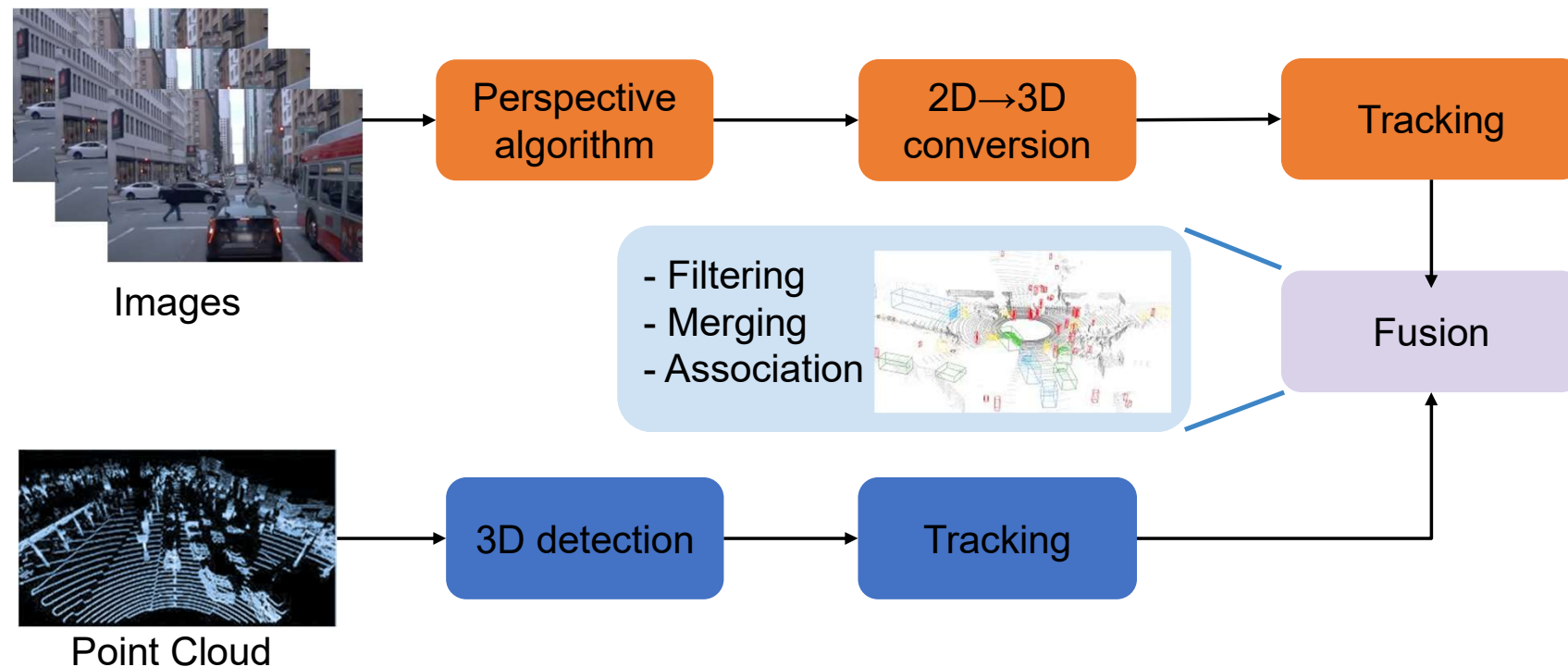
L9

ET5

ES7

P7

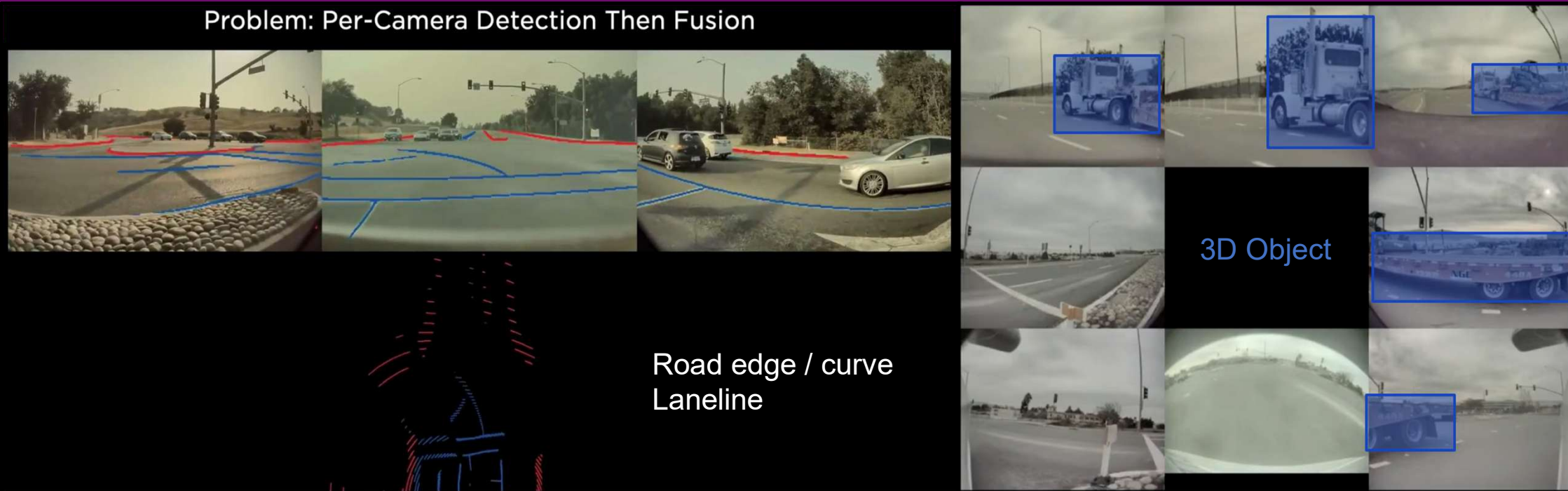
Per-camera Detection then Fusion



- Different networks for different sub-tasks in perspective view
- Fuse perception results based on many rules and priors

Per-camera Detection then Fusion

Problem: Per-Camera Detection Then Fusion



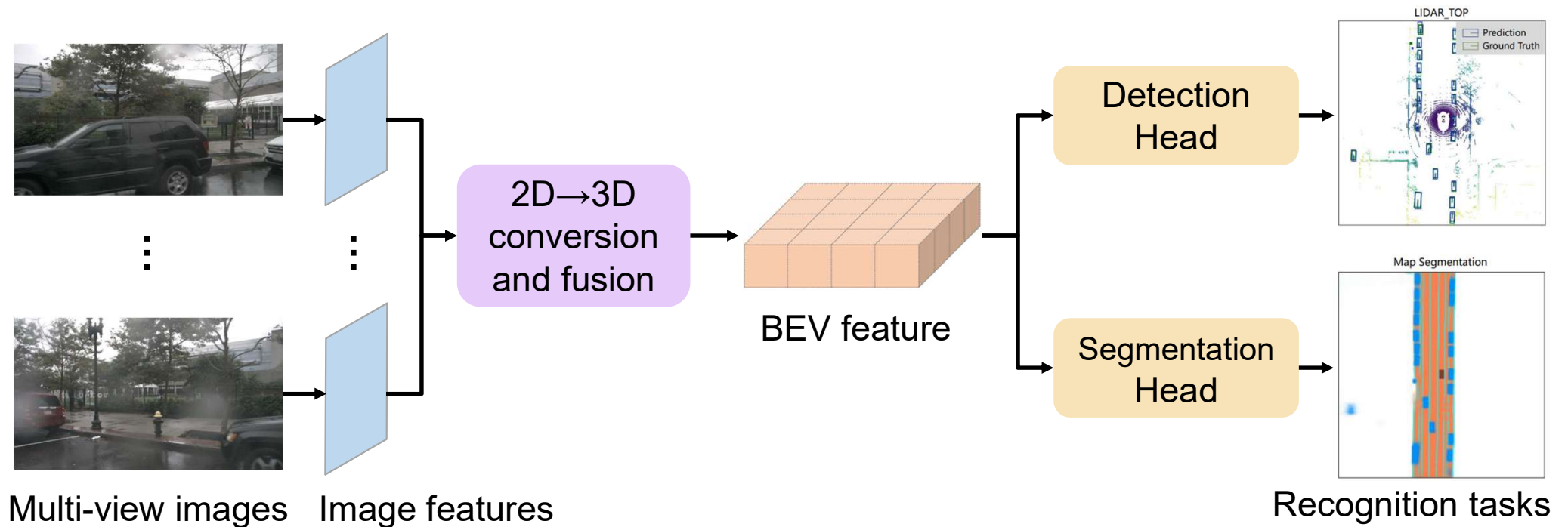
Difficulty 1

- Assuming ground is horizontal ——— Not true
- Project from image plane to BEV space ——— No depth per pixel

Difficulty 2

Fusion is difficult as objects span across images

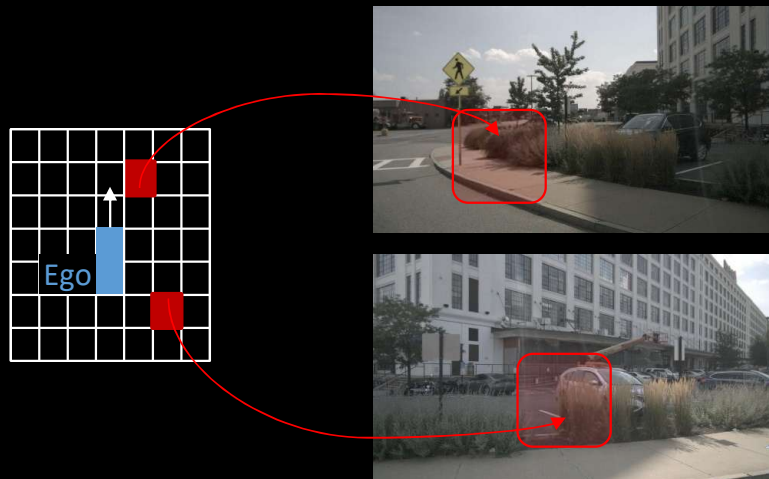
Bird's-Eye-View (BEV) Recognition



BEV recognition: construct unified BEV feature fusing multi-view features, and handle different recognition tasks based on the BEV feature

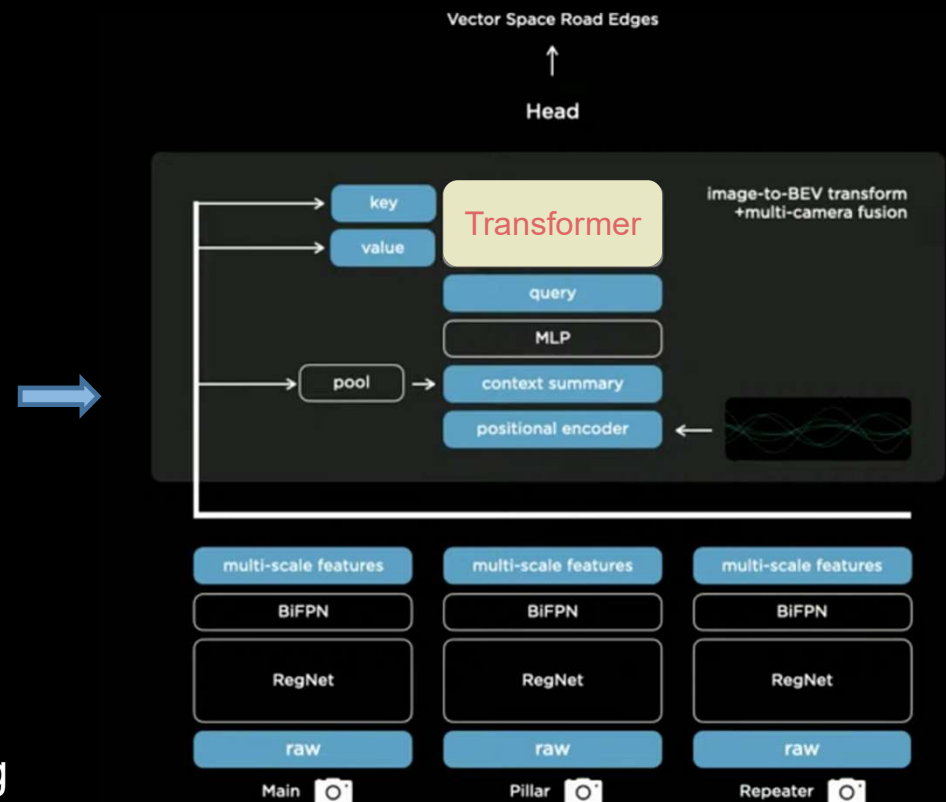
BEV-based End-to-End Perception

Where should a BEV grid look?
Approximate projection by camera calibration?



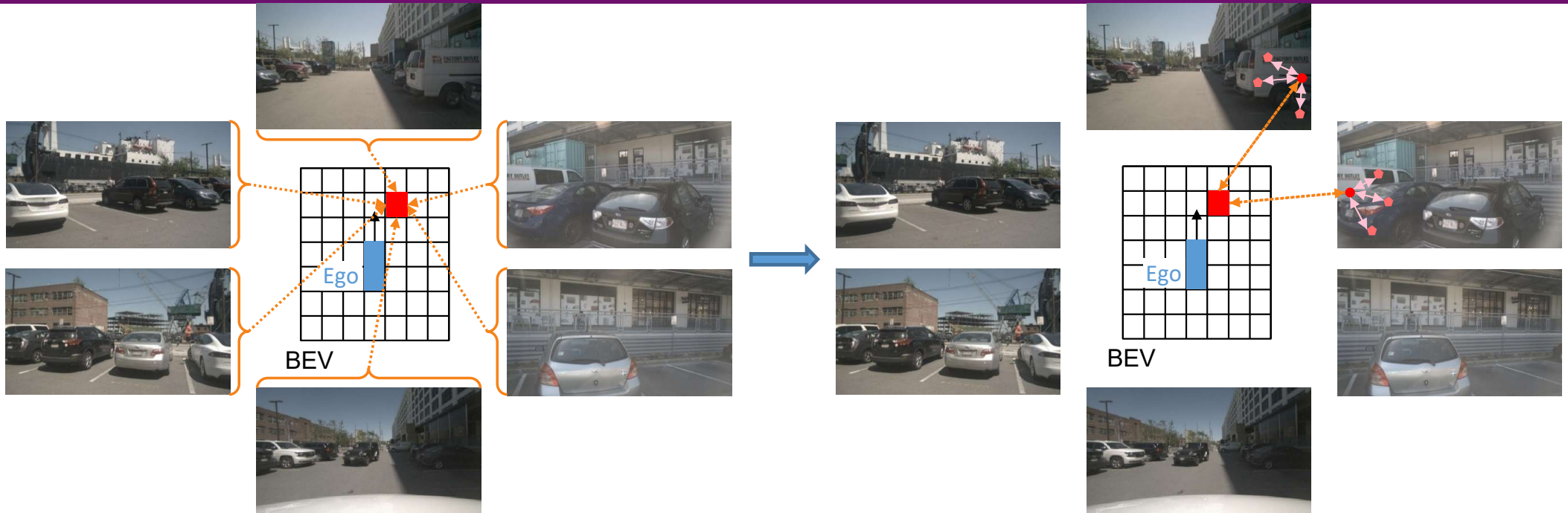
- Because of the geometry of road, projection cannot precisely project corresponding point to BEV
- If some part is occluded, the projection will be wrong

Need to find **relationship** between BEV grids and image patches.



Solution: Transformer

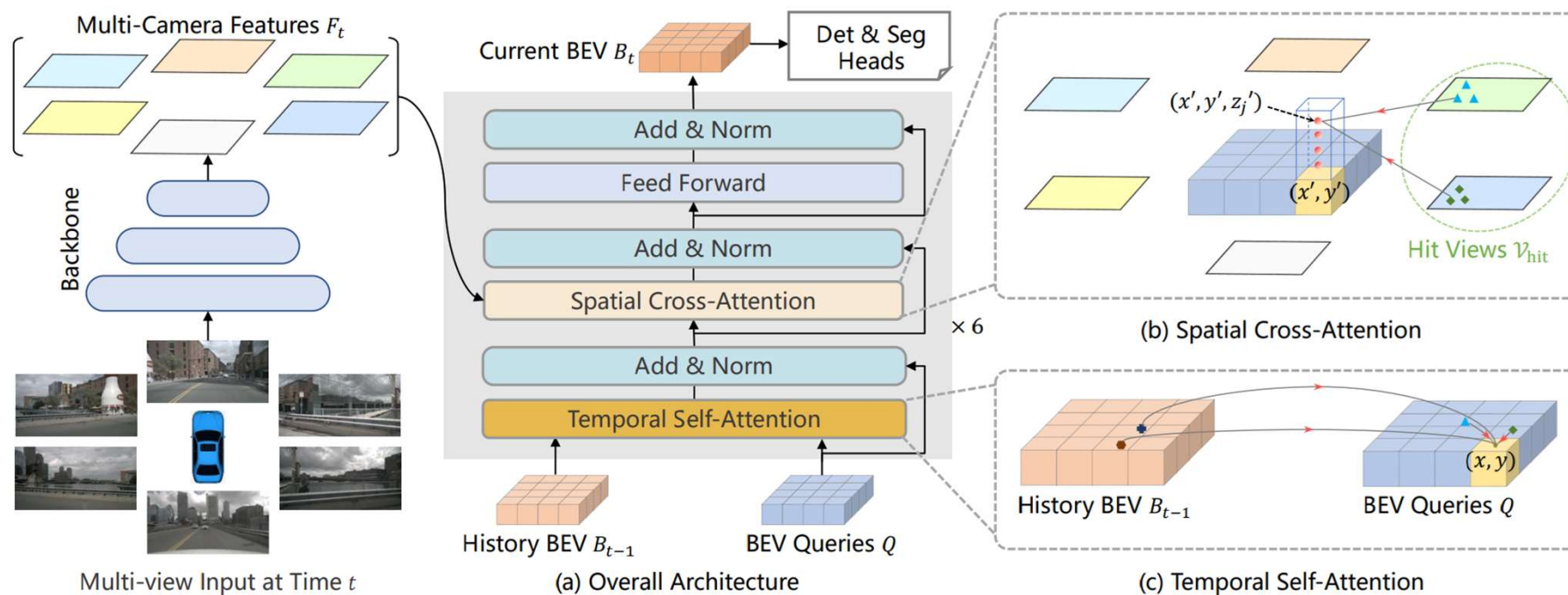
Global Attention -> Deformable Attention



Global attention: consider **all features** with **dense sampling**, suffers from **expensive** computational and memory costs

Deformable attention: focus on **RoIs** around **projected reference points** with **sparse sampling**, which is **effective**

BEVFormer: Overall Architecture



Key designs

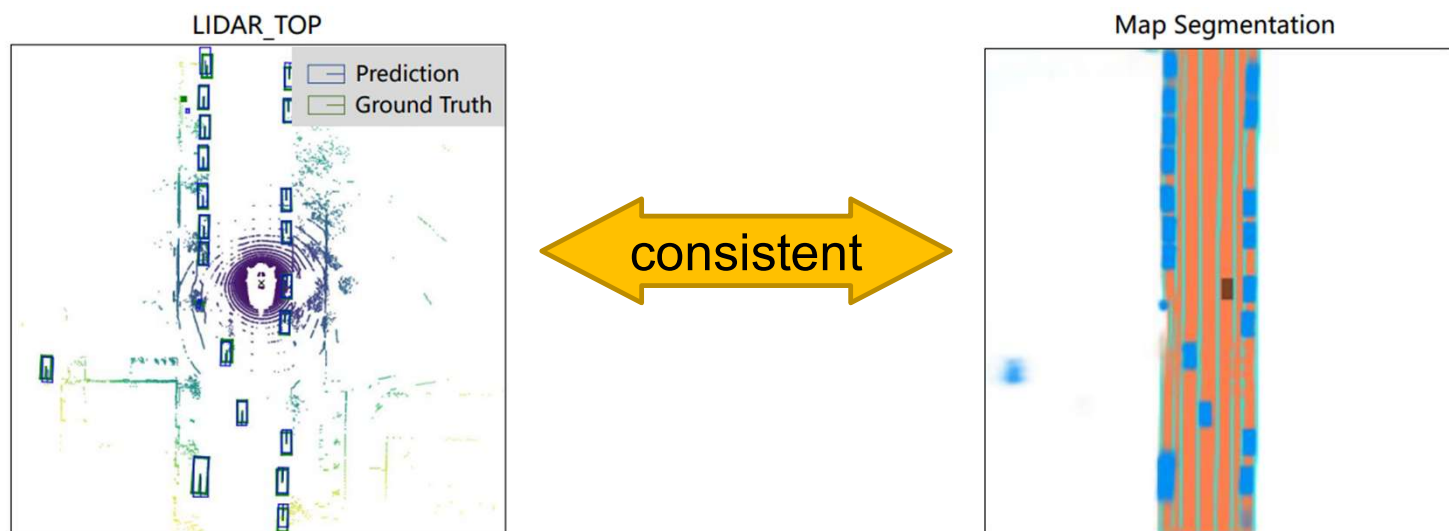
- Learnable BEV queries: query features in BEV space
- Spatial Cross-Attention: fuse multi-view features
- Temporal Self-Attention: fuse temporal BEV features

Support Multiple Perception Tasks

BEV features is **pseudo-2D features**, so 2D perception heads can be directly used:

- **Map Segmentation**: mask decoder of Panoptic SegFormer
- **3D Object Detection**: Deformable DETR

With the shared BEV features, the results of map segmentation and 3D object detection are consistent.

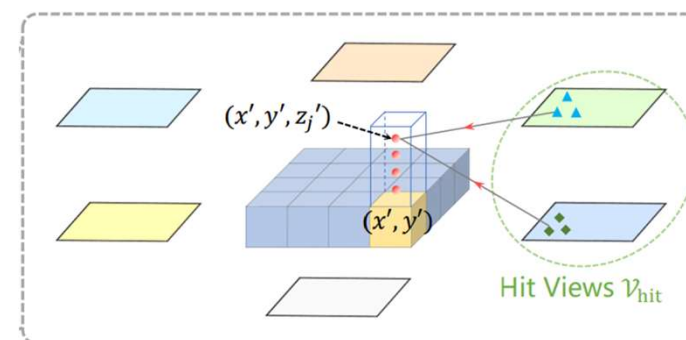


Exp: Deformable Attention is Better

Table 5: **The detection results on nuScenes val set. Comparison of different types of spatial cross-attention.** “Memory” is the consumed GPU memory during training. †: Use fp16 weights, single-scale image feature, and change the spatial shape of BEV queries from 200×200 to 100×100 .

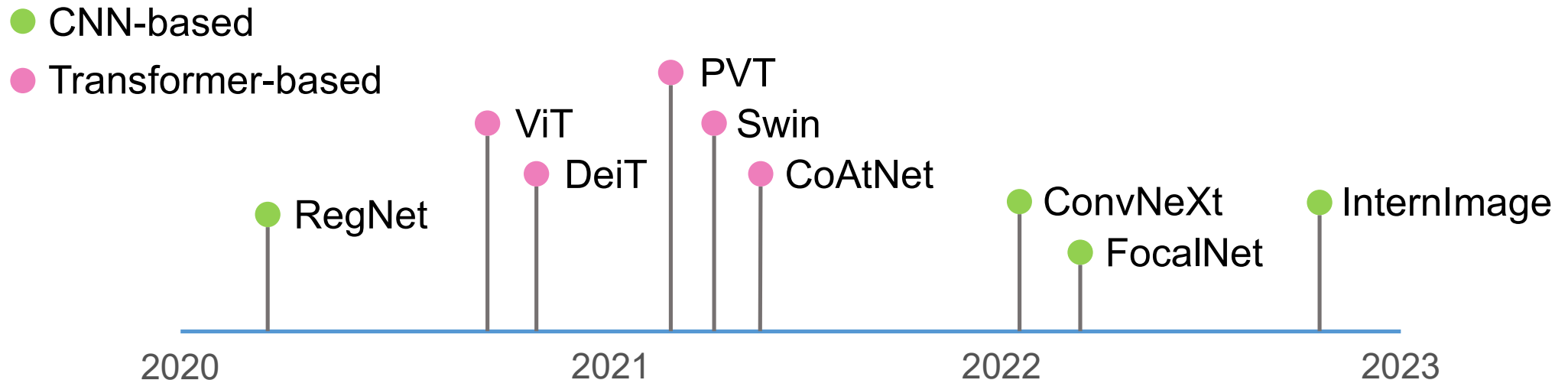
Method	Attention	NDS↑	mAP↑	mATE↓	mAOE↓	#Param.	FLOPs	Memory
BEVFormer-S [†]	Global	0.404	0.325	0.837	0.442	62.1M	1245.1G	~36G
BEVFormer-S	Deformable	0.448	0.375	0.725	0.391	68.7M	1303.5G	~20G

- Global and Dense attention is **too expensive**
- We use the deformable attention **balancing the computational cost and receptive field**



(b) Spatial Cross-Attention

Modern Image Backbones



Novel architectures and large scales
High performance on 2D recognition tasks



Unleash their power on BEV recognition

Problem:

Trained on general 2D vision tasks
Lack of specific 3D knowledge

Adapt 2D Backbone to BEV Recognition



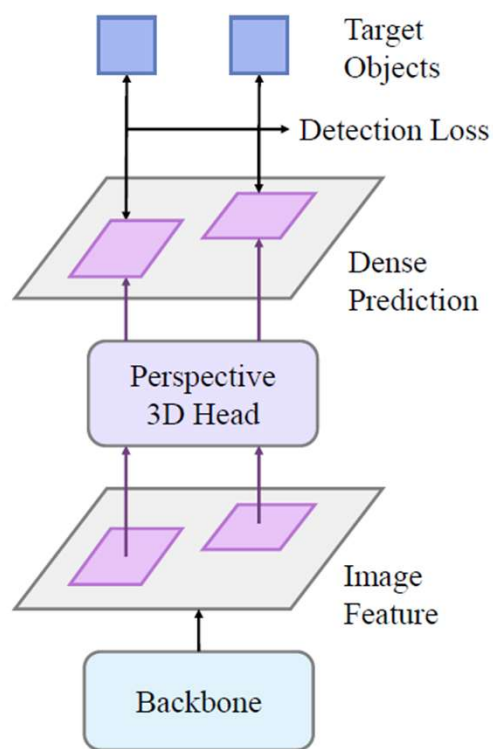
Challenge 1:

Domain gap between natural images and autonomous driving scenes

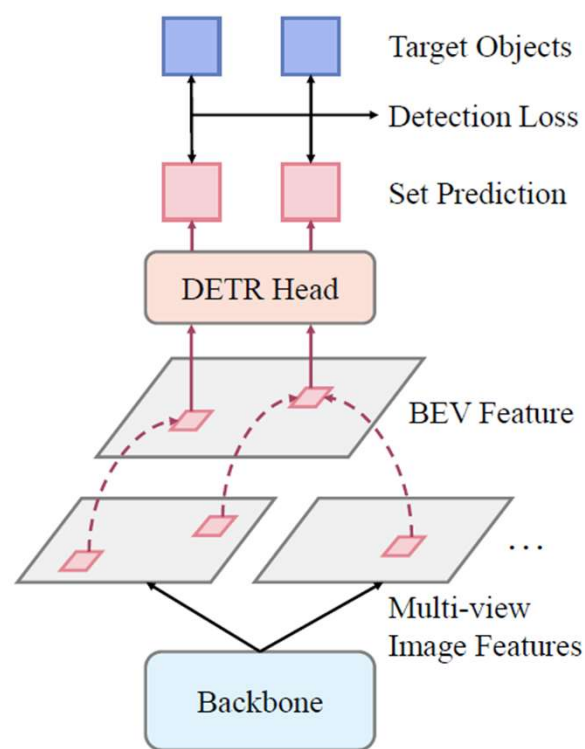
Challenge 2:

Optimization problem of BEV detectors: complex structure, indirect supervision

BEV Supervision vs. Perspective Supervision



(a) Perspective Supervision



(b) BEV Supervision

BEV Supervision:

Indirect to the backbone

Insufficient guidance

Perspective Supervision:

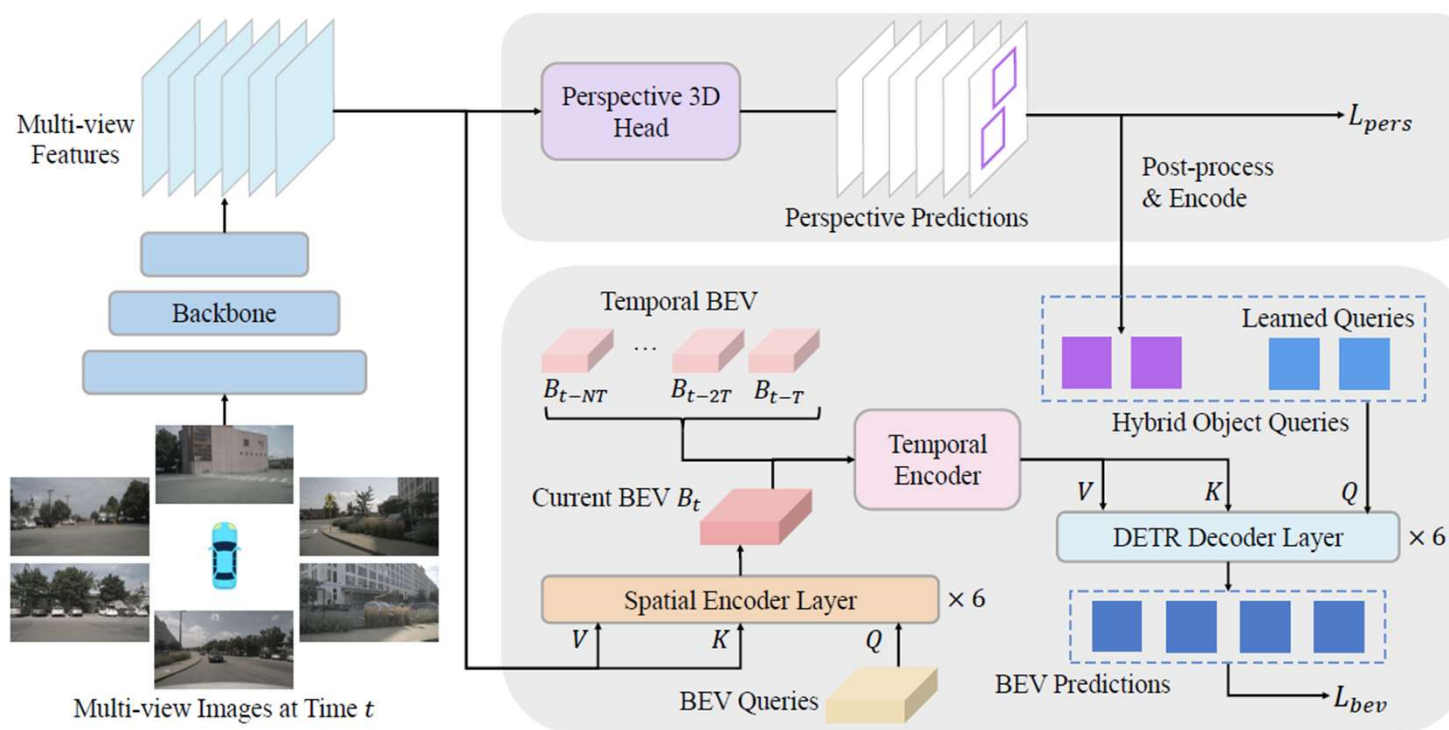
Direct to the backbone

explicitly guidance to perceive 3D scenes

Our solution:

An **extra perspective head** for **auxiliary perspective loss**

BEVFormer V2: Overall Architecture



Key designs

- Perspective supervision: facilitate the optimization
- Hybrid object query: construct two-stage pipeline

Exp: Perspective Supervision Ablation

Table 3. The results of perspective supervision with different 2D image backbones on the nuScenes *val* set. ‘BEV Only’ and ‘Perspective & BEV’ are the same as Tab. 2. All the backbones are initialized with COCO [20] pretrained weights and all models are trained without temporal information.

Backbone	Epoch	View Supervision	NDS	mAP	mATE	mASE	mAOE	mAVE	mAAE
ResNet-50	48	BEV Only	0.400	0.327	0.795	0.277	0.479	0.871	0.210
ResNet-50	48	Perspective & BEV	0.428	0.349	0.750	0.276	0.424	0.817	0.193
ResNet-101	48	BEV Only	0.426	0.355	0.751	0.275	0.429	0.847	0.215
ResNet-101	48	Perspective & BEV	0.451	0.374	0.730	0.270	0.379	0.773	0.205
VoVNet-99	48	BEV Only	0.441	0.367	0.734	0.271	0.402	0.815	0.205
VoVNet-99	48	Perspective & BEV	0.467	0.396	0.709	0.274	0.368	0.768	0.196
InternImage-B	48	BEV Only	0.455	0.398	0.712	0.283	0.411	0.826	0.204
InternImage-B	48	Perspective & BEV	0.485	0.417	0.696	0.275	0.354	0.734	0.182

Generalization: ~3.0 NDS and ~2.0 mAP improvement for all backbones

Better perceive 3D scenes: errors of translation (mATE), orientation (mAOE), and velocity (mAVE) are significantly lower

Impact

BEVFormer Series: top performing algorithm for bird-eye-view perception, widely recognized

- Top-10 most influential papers of ECCV 2022
- Winner of Waymo Challenge 2022

TABLE 1: Most Influential ECCV Papers (2023-01)

[BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images Via Spatiotemporal Transformers](#)

[IF:3](#) [Literature Review](#) [Related Patents](#) [Related Grants](#) [Related Orgs](#) [Related](#)

2022 **6** [Experts](#) [Details](#)

ZHIQI LI et. al.

Highlight: In this work, we present a new framework termed BEVFormer, which learns unified BEV representations with spatiotemporal transformers to support multiple autonomous driving perception tasks.

Li et. al., BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers. ECCV 2022.
Yang et. al., BEVFormer v2: Adapting Modern Image Backbones to Bird 's-Eye-View Recognition via Perspective Supervision. CVPR 2023. (Highlight)



Method Name	Object Type	Camera	Frames [-p]	LET-3D-APL	LET-3D-AP	Date (Pacific Daylight Time)
1 BEVFormer-Shanghai AI Lab	ALL_NS	ALL	0	0.5616	0.7069	2022-05-23 23:21
2 BEVFormer++	ALL_NS	ALL	0	0.5391	0.6894	2022-05-22 02:22
3 MV-FCOS3D++	ALL_NS	ALL	0	0.5127	0.6627	2022-05-24 00:04
4 FCOS3D-MVDet3D	ALL_NS	ALL	0	0.5106	0.6603	2022-05-23 22:51
5 3DMVT	ALL_NS	ALL	0	0.4876	0.6550	2022-05-23 21:00
6 CMKD	ALL_NS	ALL	0	0.3790	0.5101	2022-05-23 22:53
7 Anonymous Submission	ALL_NS	ALL	0	0.3576	0.5162	2022-05-23 06:12
8 CenterLS	ALL_NS	ALL	0	0.2914	0.4117	2022-05-23 01:53
9 MViMagnet2PointsDet	ALL_NS	ALL	0	0.2387	0.3853	2022-05-23 08:39
10 MonoWatch	ALL_NS	ALL	0	0.2163	0.2833	2022-05-23 22:15
11 lotCV3dDet	ALL_NS	ALL	0	0.1670	0.2252	2022-05-21 23:49
12 LeapMotor_MonoDet	ALL_NS	ALL	0	0.1652	0.2387	2022-05-23 00:14

BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers

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